

Soft Spaces - Phase 3.2

Checkpoint: Global $\pm E$ Symmetry and Local Hotspot Mechanism

Working checkpoint - 10 September 2026

Purpose. This checkpoint records the current Phase 3.2 result chain before further mechanism work. It separates what is numerically established from what remains a working law. The central outcome is that the global $\pm E$ organization is algebraically explained, while local hotspot prominence is statistically linked to excess constructive coherence among the virtual $P \rightarrow Q \rightarrow P$ pathways.

1. Frozen Phase 3.2 setting

Item	Frozen choice
Hilbert space	8 qubits, dimension 256
Hamiltonian	Frozen 11-term random-Pauli model inherited from Phase 3.1
Seeds	25042000–25042011
Eigenspaces	Complete degeneracy-closed P subspaces; 64 groups of dimension 4 for the reference seed
Perturbation families	Dephasing Z/ZZ and transverse X/XX ; five perturbation terms
Regularization	$g_\eta(x) = \frac{x}{x^2 + \eta^2}$, with $\eta = 10^{-3}$
Primary response	Frozen Phase-3.1 hotspot prominence

2. Global layer: the $\pm E$ structure is explained

The decisive Phase 3.2 symmetry result is the existence, for the reference frozen Hamiltonian, of a Pauli-string generator

$$\Gamma = \text{XIXXIIXX},$$

with the exact anticommutation relation

$$\boxed{\Gamma H \Gamma^\dagger = -H.}$$

The perturbations transform covariantly,

$$\boxed{\Gamma V \Gamma^\dagger = sV}, \quad s = \begin{cases} -1, & \text{dephasing,} \\ +1, & \text{transverse.} \end{cases}$$

This induces the projector mappings

$$P_{-E} = \Gamma P_E \Gamma^\dagger, \quad Q_{-E} = \Gamma Q_E \Gamma^\dagger.$$

For the regularized second-order effective operator

$$\Sigma_E = P_E V Q_E g_\eta(E - Q_E H Q_E) Q_E V P_E,$$

the sign is fixed:

$$\boxed{\Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger.}$$

The v32.2 residuals for the required mapping chain were of order 10^{-12} or below, while the deliberately wrong plus-sign relation produced residual 2.0. Consequently,

$$\|\Sigma_{-E}\|_F = \|\Sigma_E\|_F,$$

and the corresponding $\pm E$ Frobenius-norm symmetry follows from algebraic covariance rather than from an empirical coincidence.

3. What v32.3–v32.7 ruled in and ruled out

Version	Test	Main conclusion
v32.3	Reference-seed local scan	Σ -based descriptors correlated with prominence; transverse looked strong, but the result was seed-specific.
v32.4	Across-seed confirmation	Transverse <code>real_sigma</code> median Spearman +0.4222; no stable single-scalar law.
v32.5	Multivariate LOSO	Transverse pooled LOSO $\rho = +0.2991$, $R^2 = +0.0693$; positive fold rank correlation in 12/12 seeds.
v32.6	Weighted Q spectrum	Transverse pooled LOSO $\rho = +0.2571$, $R^2 = -0.0578$; simple Q -density/gap descriptors did not supply the missing variable.
v32.7	Internal matrix shape of Σ	Transverse pooled LOSO $\rho = +0.2846$, $R^2 = -0.2352$; static anisotropy/rank was insufficient.

4. v32.8–v32.9: interference/coherence emerges

The key conceptual shift was to stop treating Σ_E only as a final operator and instead decompose it into individual virtual Q -state contributions,

$$\Sigma_E = \sum_{q \in Q} T_q, \quad T_q = g_\eta(E - E_q) PV|q\rangle\langle q|VP.$$

This motivates the dimensionless coherence factor

$$C_E = \frac{\|\sum_q T_q\|_F}{\sum_q \|T_q\|_F}.$$

C_E is high when the virtual pathways reinforce one another and low when they cancel strongly.

Family/metric	Median Spearman	Consistency
Dephasing C_E	+0.3088	11/12 positive
Transverse C_E	+0.5423	11/12 positive
Transverse positive/negative balance	+0.4987	12/12 positive
Transverse scalar cancellation	−0.4987	12/12 negative

5. v32.10–v32.11: intervention against a norm-preserving null

To test whether the association was genuinely about alignment/cancellation rather than merely about the magnitudes of the individual pathways, the signs of the T_q contributions were randomized independently while every individual norm $\|T_q\|_F$ was held fixed.

The primary endpoint was the coherence excess

$$H_E = C_E - \langle C_E \rangle_{\text{null}}.$$

The strict v32.11 confirmation used 1000 norm-preserving sign randomizations per pair together with seed-level permutation tests.

Family	Median seed ρ	Positive seeds	Sign-test p	Pooled ρ	Gate
Dephasing	+0.3292	11/12	0.00317	+0.3039	5/5 - STRONG
Transverse	+0.3713	10/12	0.01929	+0.3068	5/5 - STRONG

Interpretation: after the individual pathway norms are held fixed, excess coherence above the randomized alignment null still tracks hotspot prominence. This is stronger evidence than a static correlation, although it remains a computational intervention rather than an analytic derivation.

6. v32.12: provisional local hotspot law

Phase 3.2 can now state a compact empirical local-law parameter:

$$H_E = C_E - \langle C_E \rangle_{\text{null}}, \quad C_E = \frac{\left\| \sum_q T_q \right\|_F}{\sum_q \|T_q\|_F}.$$

The tested statement is monotonic rather than an exact amplitude equation:

$$H_E \uparrow \implies \text{hotspot prominence}(E) \text{ increases statistically.}$$

Family	Median seed $\rho(H_E)$	Positive seeds	Sign-test p	Pooled ρ	Median H_E
Dephasing	+0.3294	12/12	0.00024	+0.3029	+0.01710
Transverse	+0.3630	10/12	0.01929	+0.3054	+0.01974

The strict TOP/MIDDLE/BOTTOM ordering is not universal: it is monotone in 5/12 dephasing seeds and 4/12 transverse seeds. Therefore H_E is currently a statistically reproducible ranking parameter, not a deterministic per-hotspot law.

7. Current two-layer Phase 3.2 picture

Global layer

$$\Gamma H \Gamma^\dagger = -H \implies E \leftrightarrow -E \implies P_{-E} = \Gamma P_E \Gamma^\dagger, \quad \Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger.$$

Local layer

$$H_E = C_E - \langle C_E \rangle_{\text{null}}, \quad H_E \uparrow \iff \text{prominence tends to increase.}$$

The first layer explains why $\pm E$ partners are organized symmetrically. The second layer is the current mechanism candidate for why some of those symmetry-related pairs become stronger hotspots than others.

8. Established, supported, and still open

Status	Statement
Established for reference symmetry chain	Exact Γ anticommutation; $\pm E$ projector mapping; fixed minus sign for Σ covariance; Frobenius-norm symmetry follows algebraically.
Strongly supported across 12 seeds	Interference/coherence descriptors track prominence; norm-preserving randomization leaves a significant coherence-excess signal; H_E passes the current cross-seed gates.
Not established	A closed analytic equation predicting numerical prominence amplitude; universal strict TOP/MID/BOTTOM ordering; proof that the same explicit Γ string applies unchanged to every seed.
Next scientific target	Derive or constrain a mapping such as $\text{prominence} = f(H_E, \dots)$ from operator perturbation structure, or identify the minimal additional variable needed for amplitude prediction.

9. Checkpoint conclusion

Phase 3.2 has moved from an empirical observation of $\pm E$ organization to a coherent two-layer mechanism. The global layer is algebraic: an anticommuting symmetry organizes the spectrum and the effective second-order operator. The local layer is represented by excess pathway coherence H_E : hotspots are statistically stronger where the actual virtual Q pathways add more coherently than in a norm-preserving randomized reference.

This is sufficiently stable to freeze as a checkpoint, but it is not yet sufficient to claim a complete analytic hotspot law.